

Cryptography and PKI

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Agenda



- **Introduction to Cryptography**
 - Substitution Ciphers
 - Transposition Ciphers
 - Block and Stream Ciphers
 - DES
- **Hash Functions**
- **Symmetric Key Cryptography**
- **Asymmetric key Cryptography**

Information security properties



- General definition: Information security involves providing appropriate levels of assurance of

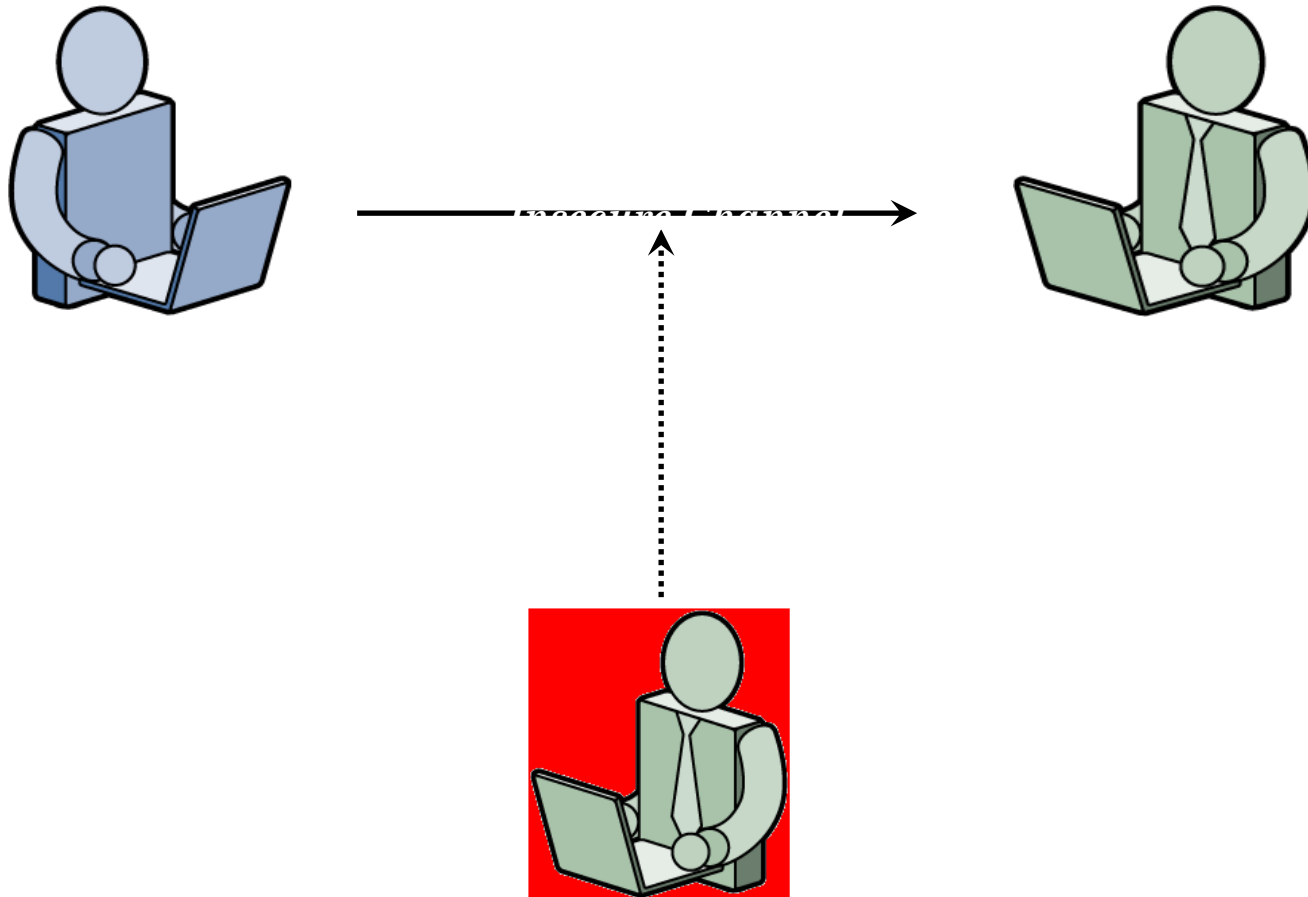
Privacy/Confidentiality: preventing disclosure of information to unauthorized individuals or systems

Authenticity: Ensuring that the user, data, transactions, communications or documents are genuine

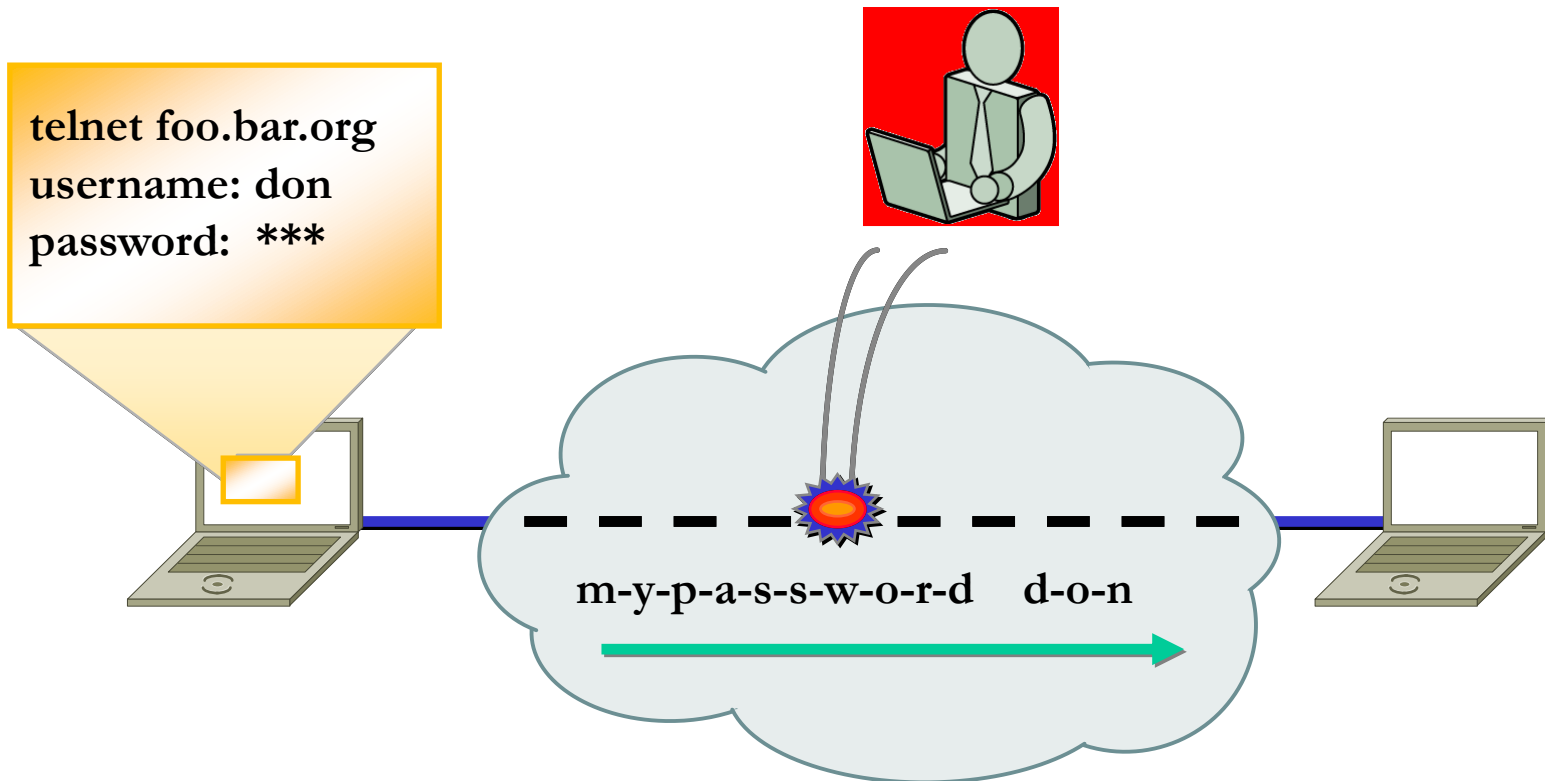
Integrity : Data cannot be modified without authorization

Non-Repudiability: One party of a transaction can not deny having sent/received a transaction

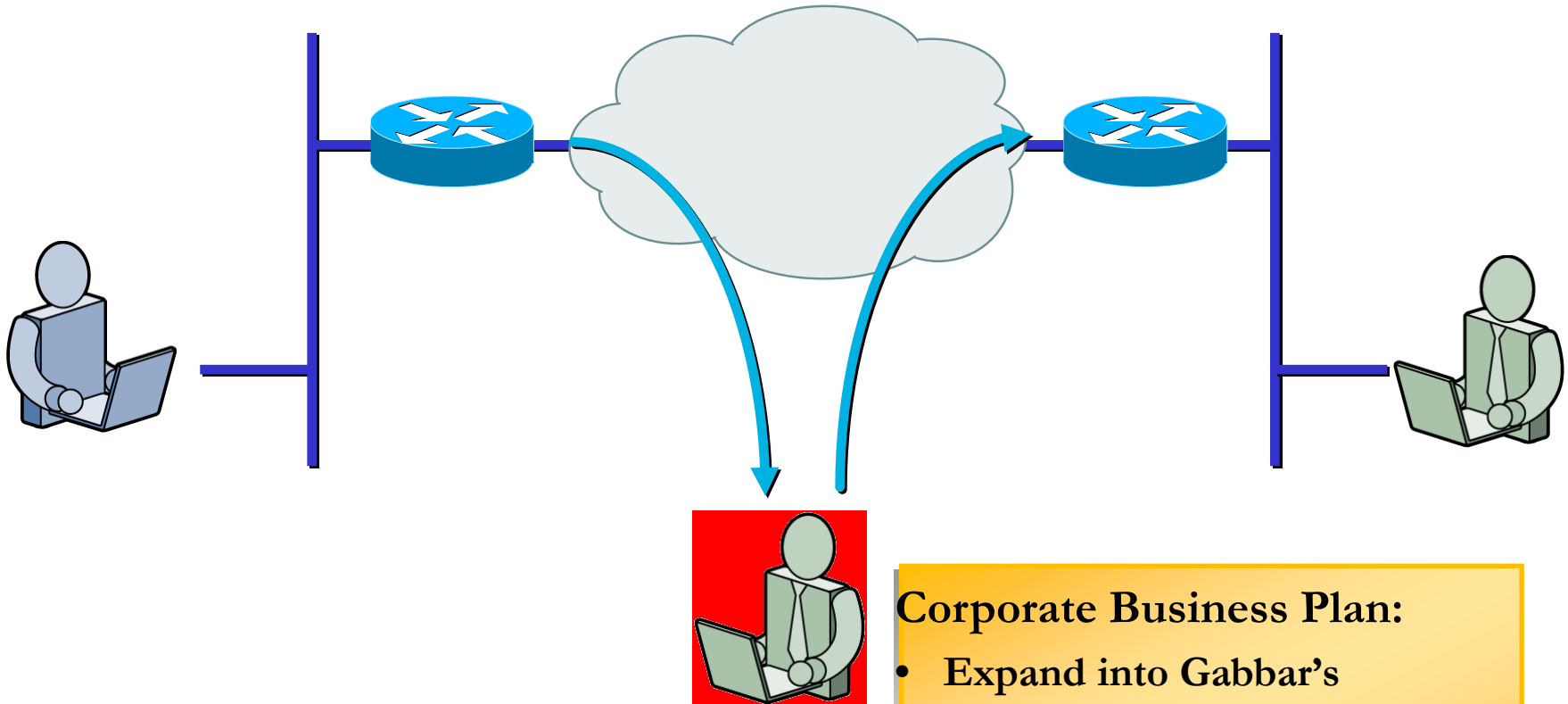
The Scenario



Threats: Packet Sniffing



Threats: Data Theft



Corporate Business Plan:

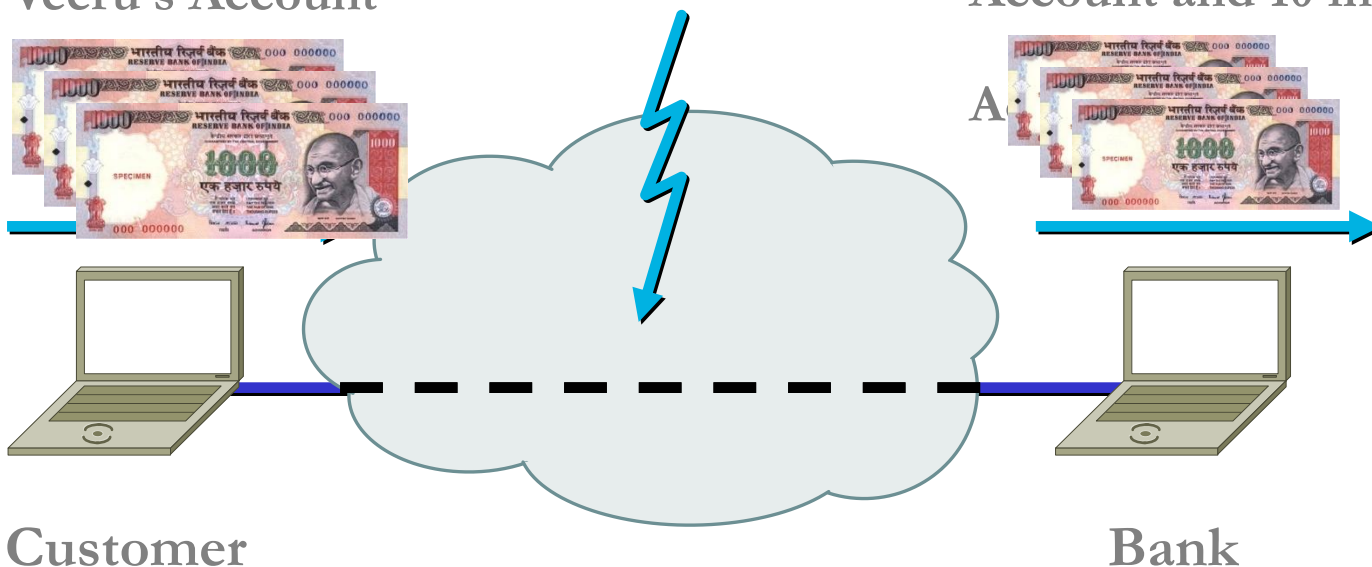
- Expand into Gabbar's core area
- Massively discount our products for next quarter

Threats: Data Alteration

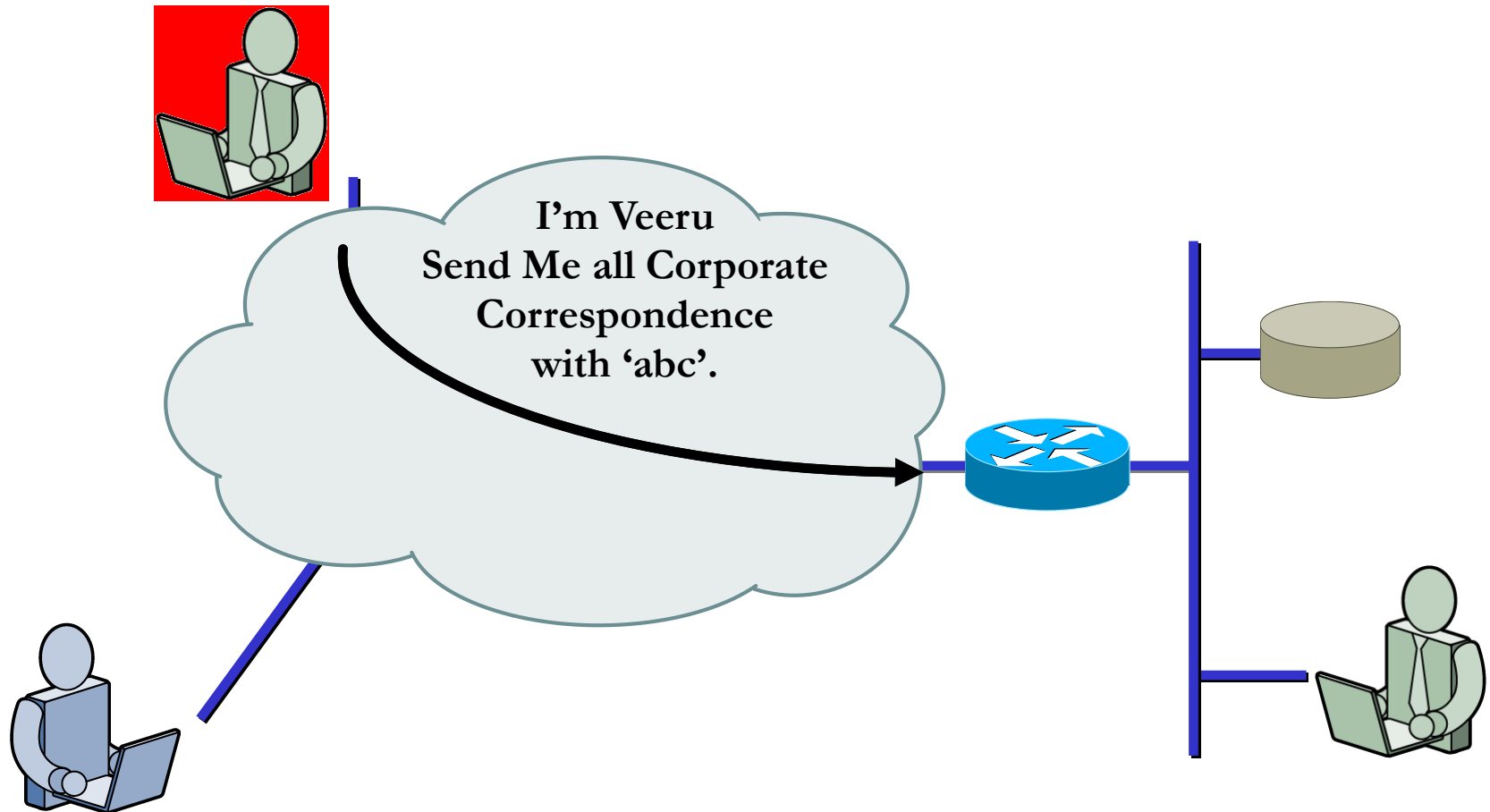
Deposit 1,00,000
in Veeru's Account



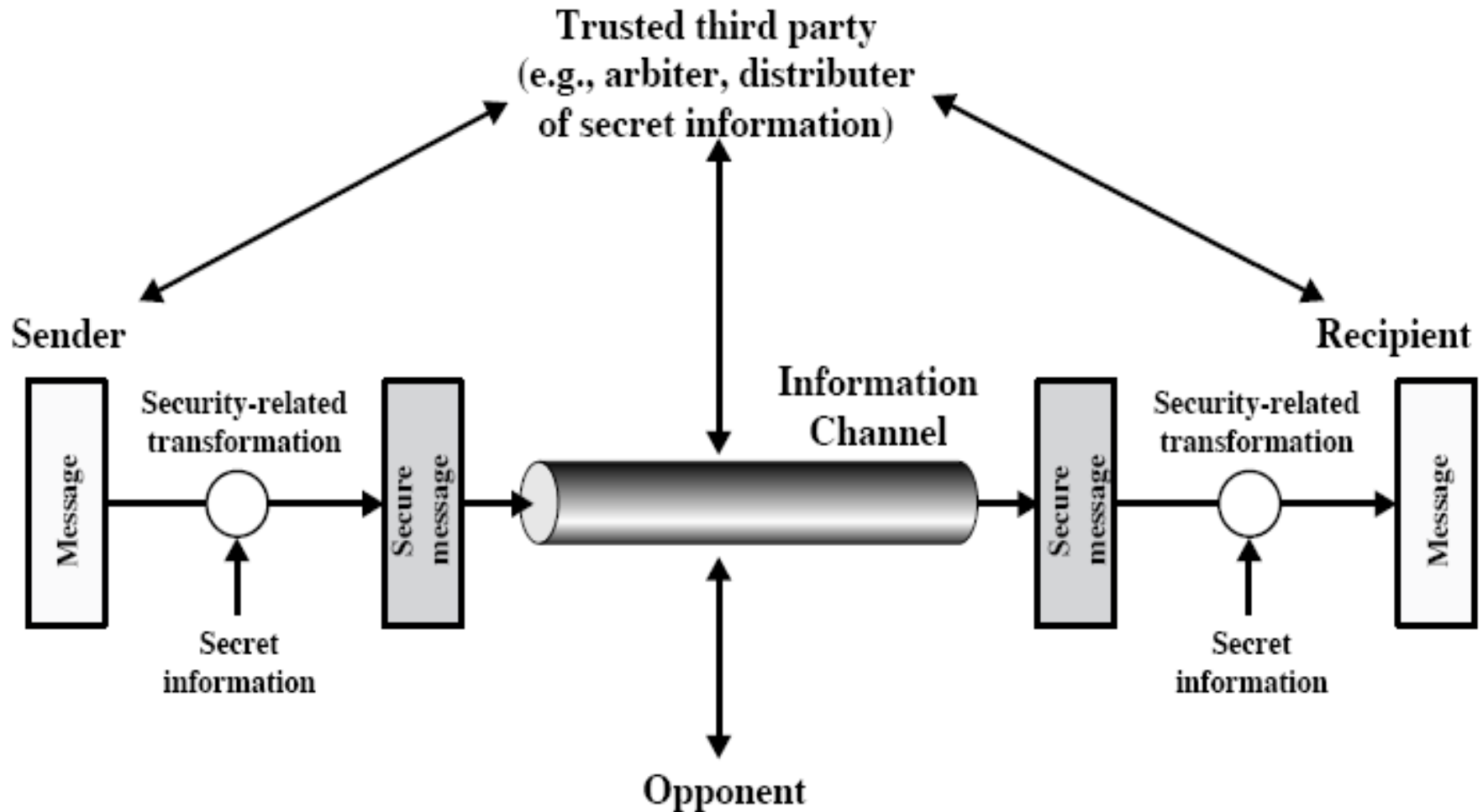
Deposit 99,990 in Gabbar's
Account and 10 in Veeru's



Threats: Spoofing



Model for Network Security



Security Threats

- Threats can come from a range of sources
- Various surveys, with results of order:
 - 55% human error
 - 10% disgruntled employees
 - 10% dishonest employees
 - 10% outsider access
 - also have "acts of god" (fire, flood etc)

List of some common attacks:

- Replay Attack
- Insider Attack
- Stolen Verifier Attack
- Shoulder Surfing Attack
- Spoofing Attack
- Guessing Attack
- Dictionary Attack
- Brute-force Attack
- Man-In-The-Middle (MITM)
- Phishing Attack

- Cryptosystems can be classified based on
 - Type of operations - Substitution/Transposition
 - Way in which plaintext is processed - Block/Stream
 - Number of keys used - Symmetric / Asymmetric
 - Hash Functions
 - Steganography

Cryptography Terminology



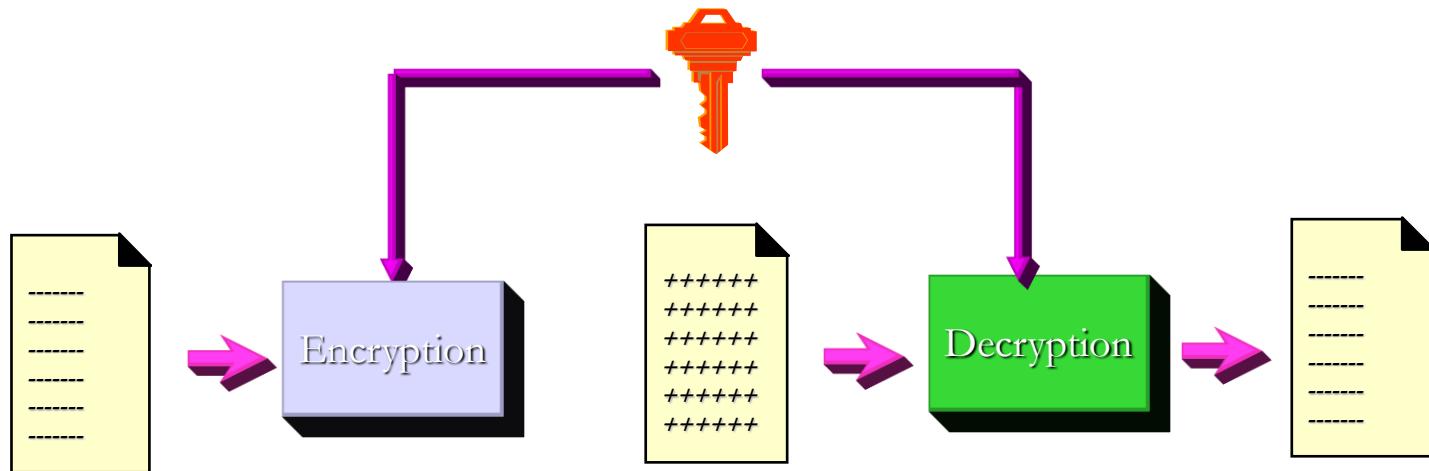
- **Cryptography:** The study & practice of hiding, encrypting or secret writing using mathematical & logical principles to secure information.
- **Cryptanalysis:** The study of ciphers, cipher text, or cryptosystems with a view to finding weaknesses in them that will permit retrieval of the plaintext from the cipher text, without necessarily knowing the key or the algorithm.
- **Plaintext:** The message which has to be sent to other party.
- **Encryption / Decryption:** The process of transforming plain text input to an un-interpretable form is called Encryption. Decryption is reverse of Encryption. Therefore, this is a two-way function.

Cryptography ...



- **Cipher text:** The message after it is encoded
 - **Key:** This is a unique value (bit pattern, alphabetical sequence) that is used by the cipher for encryption/decryption
 - **Key space:** The range of possible values used to construct keys
 - Eg: A 4 digit key (0-9) can have a key space of 10,000
 - A 6 digits key can have a key space of 1000,000 keys
 - **Key Clustering:** When two different keys generate the *same* cipher text from the *same* plaintext
 - **Work factor:** The amount of processing power (resources and time to break a cryptosystem)
-

Encryption / Decryption



“The quick
brown fox
jumps over
the lazy dog”

“AxCv;5bmEseTfid3)fG
smWe#4^,sdgfMwir3:d
kJeTsY8R\s@!q3%”

“The quick
brown fox
jumps over
the lazy dog”

History

- Cryptography is quite old – at least about 4000 years.
- Ancient Egyptians use Symbols to represent things, an early form of writing (1900 BC)
- 1500 BC The Phoenicians developed an alphabet
- 600 BC Palestinians use the Atbash cipher
- 500 BC The Spartans use the encryption process Scytale



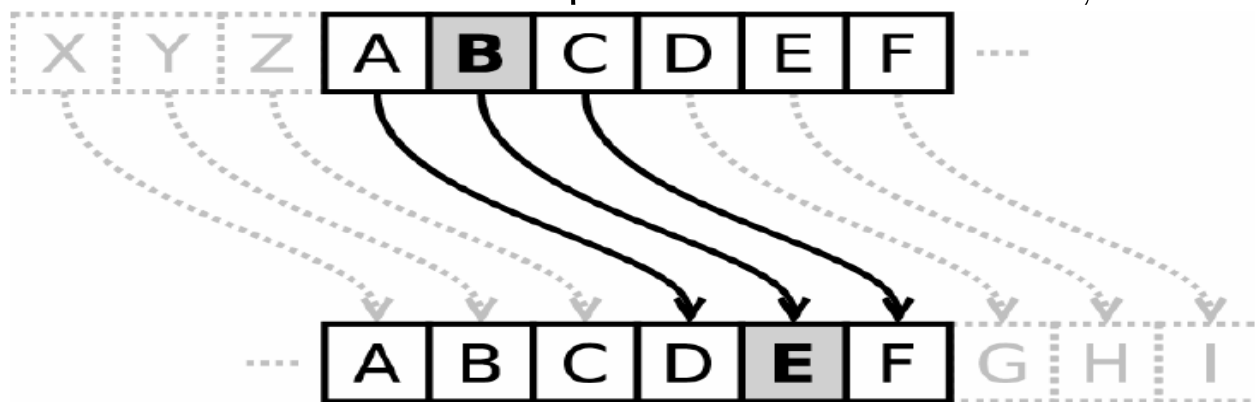
- In 50 BC, Julius Caesar used an alphabet with a shift of three and hence named as Caesar cipher.
- Blaise de Vigenère discussed Vigenere cipher in 1585 AD
- 1917 AD American, Gilbert S. Vernam, develops the One-time-pad
- 1976 AD Diffie-Hellman key exchange protocol is developed
- 1977 AD DES is developed by IBM
- 1977 RSA is developed, this method is still widely used today
- 2000 AD AES is chosen as the successor to DES

Substitution Ciphers

- Here each character is simply represented by another character

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
H	X	V	J	D	I	T	U	E	R	G	A	L	S	F	P	W	Z	M	K	Q	B	Y	O	C	N

- In its simplest form there is no logic in order of representation.
- A type of substitution cipher is Caesar Cipher (Shift cipher) where each character in cipher text is shifted by 'k' letters.



Eg: Caesar Ciphers



KRISHNA $\rightarrow$ nulvkqd obvious

Shift by k letters (here $k = 3$)

ABCDEFGHIJKLMNOPQRSTUVWXYZ

Shift by k letters (here $k = 6$)

KRISHNA $\rightarrow$ qxoyntg ... still obvious(?!)

Atbash

This cipher simply represents letters of the alphabet in reverse order: Eg:

Plaintext: abcdefghijklmnopqrstuvwxyz

Ciphertext: ZYXWVUTSRQPONMLKJIHGFEDCBA

Vigenère cipher

- Encryption process combines one character of plain text and corresponding character of Key to get a character of cipher text from Vigenere Square

Eg: Text: **SQUARE**

Key: **FROGFR**

Cipher Text: XHIGWV

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Playfair Cipher

- Makes use of diagraphs and comprises of several small steps
 - Key:** Treat your password like your toothbrush. Don't let anybody else use it, and get a new one every six months
 - TREAYOUPSWDLIKHBNGVXMCQZ
- Plain Text: “Information is not knowledge”
“IN FO RM AT IO NI SN OT KN OW LE DG EX”
- Cipher Text: LG MP TC YR DP GL UV DO LV UO IR IB YG
- “Information is not knowledge”
- = lgmptcyrdpgluvdolvuoiribyg

T	R	E	A	Y
O	U	P	S	W
D	L	I	K	H
B	N	G	V	X
M	C	F	Q	Z

*	*	*	*	*
*	A	C	B	D
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
AB => CD				

*	A	*	*	*
*	C	*	*	*
*	*	*	*	*
*	B	*	*	*
*	D	*	*	*
AB => CD				

*	*	*	*	*
*	A	*	*	C
*	*	*	*	*
*	*	*	*	*
*	D	*	*	B
AB => CD				

Transposition Ciphers



- Here the order of the character is changed

Rail Fence Cipher (*Capture fox*)

C P U E O
A T R F X

Cipher Text
CPUEOATRFX

Route Cipher (*We are discovered Flee at once*)

W R I O R F E O E
E E S V E L A N J
A D C E D E T C X

Cipher Text
EJXCTEDECDAEWRIORFEONALEVSE

Columnar Transposition (*Deposit Four Crore Rupees in our Citi Bank Account*)

K R I S H N A -- Key

D E P O S I T
F O U R C R O
R E R U P E E
S I N O U R C
I T I B A N K
A C C O U N T

Cipher Text
TOECKTSCPUAUPURNICDFRSIAIRERNNEOEITCORUOBO

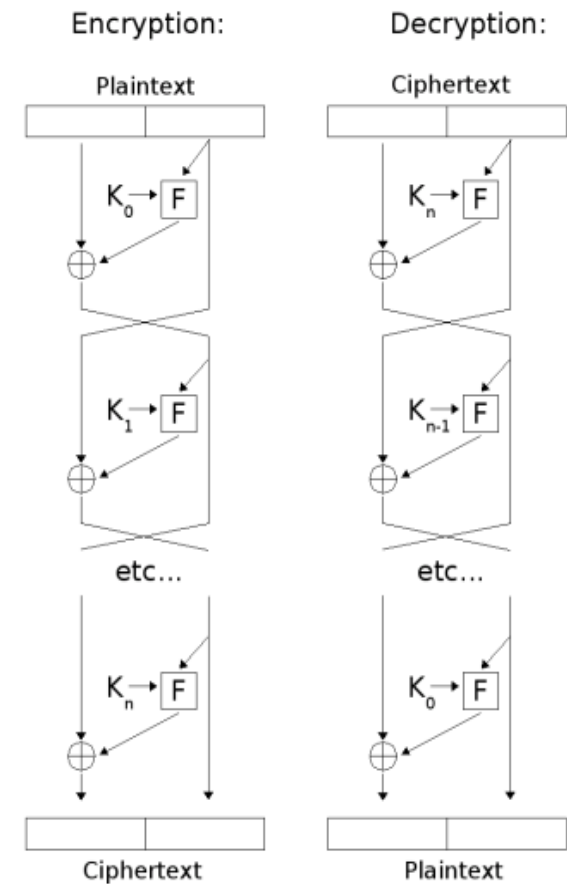
- A symmetric key cipher which operates on fixed-length groups of bits, termed *blocks*, with an unvarying transformation.
- When encrypting, a block cipher might take (for example) a 128-bit block of plaintext as input, and output a corresponding 128-bit block of ciphertext. But 64 bit is common.
- The exact transformation is controlled using a second input “the secret key.”
- Decryption is similar

Iterated block ciphers

- Most block ciphers are constructed by repeatedly applying a simpler function. This approach is known as *iterated block cipher*
- Each iteration is termed a round, and the repeated function is termed the round function; anywhere between 4 to 32 rounds are typical.

Feistel Cipher

- Describes the structure of a cipher
- A block cipher with a symmetric structure it is also commonly known as a **Feistel network**.
- A large proportion of block ciphers use the scheme, including the DES.
- It has the advantage that encryption and decryption operations are very similar, even identical in some cases, requiring only a reversal of the key.
- The size of the code or circuitry required to implement such a cipher is nearly halved.



Feistel Cipher

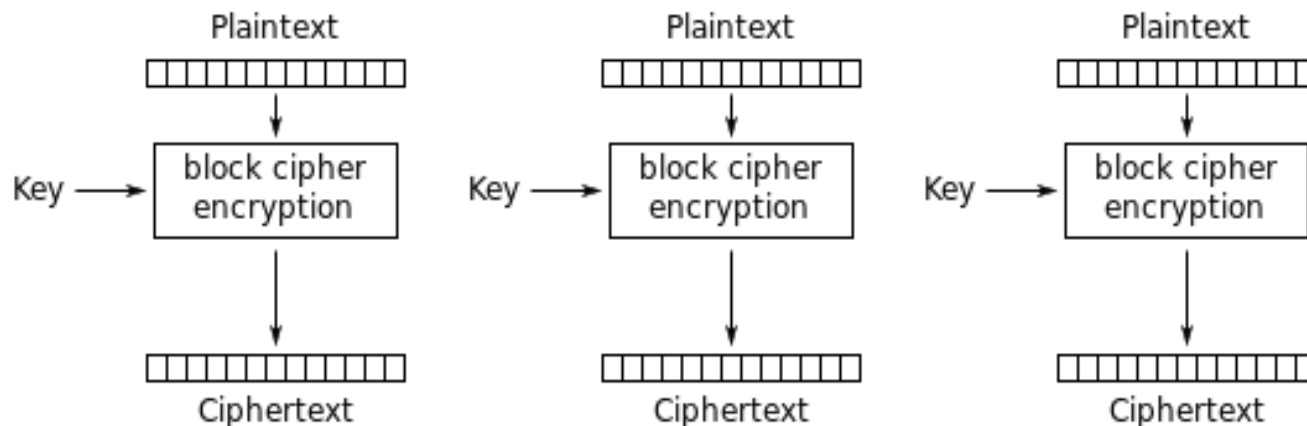
Block Ciphers



- Basis:
 - Take blocks of input and encrypt entire block
 - Reusable keys
 - Different modes
- Keep in mind potential problem areas:
 - Block padding
 - Initialization vectors
 - Codebook attacks, use the right modes

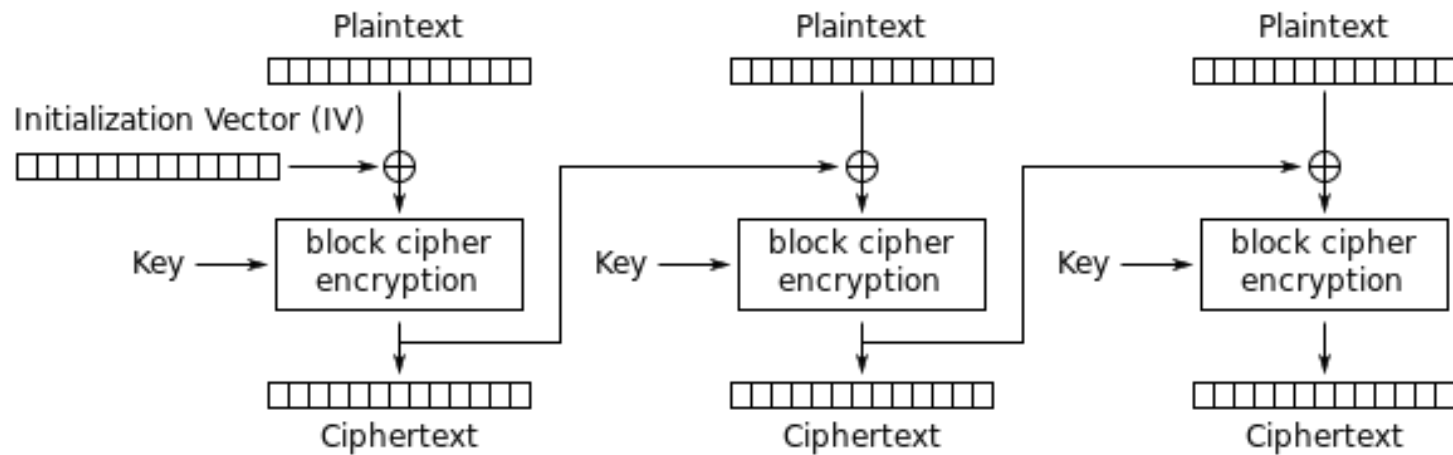
Block Ciphers

- Modes of Operation:
 - 4 Modes defined in FIPS
- **Electronic Code Book** – separately encrypt each block, patterns recognizable, “codebook” can be built up



Electronic Codebook (ECB) mode encryption

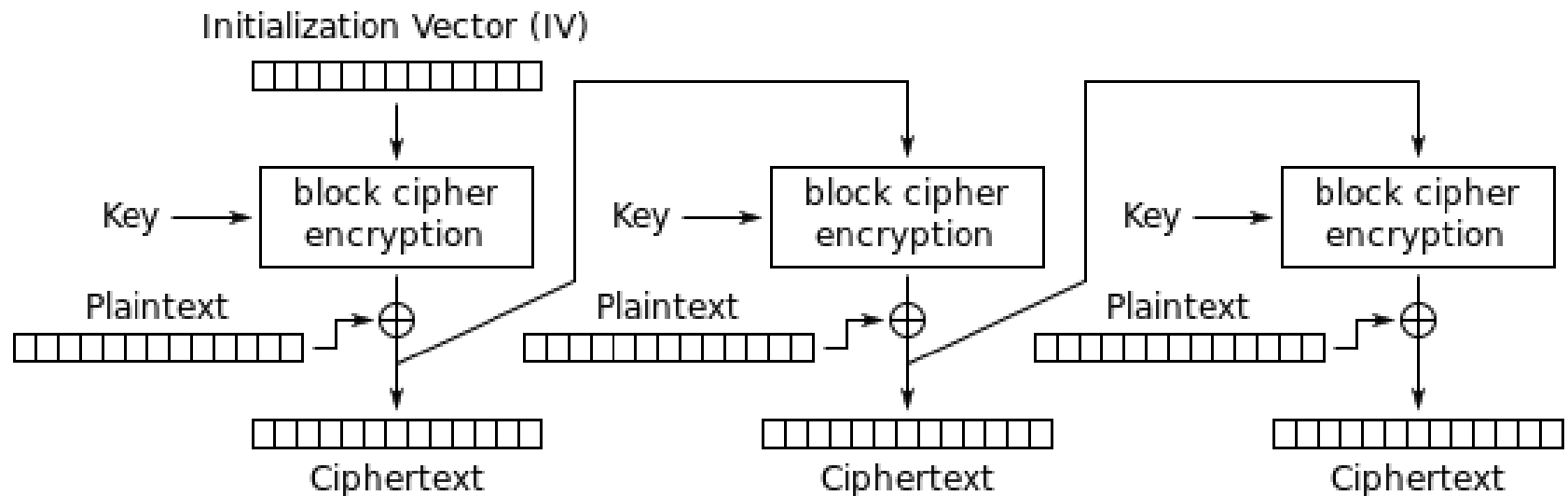
Cipher Block Chaining – XOR plaintext with previous cipher text block, then encrypt, use initialization vector for first block, makes identical inputs look different



Cipher Block Chaining (CBC) mode encryption

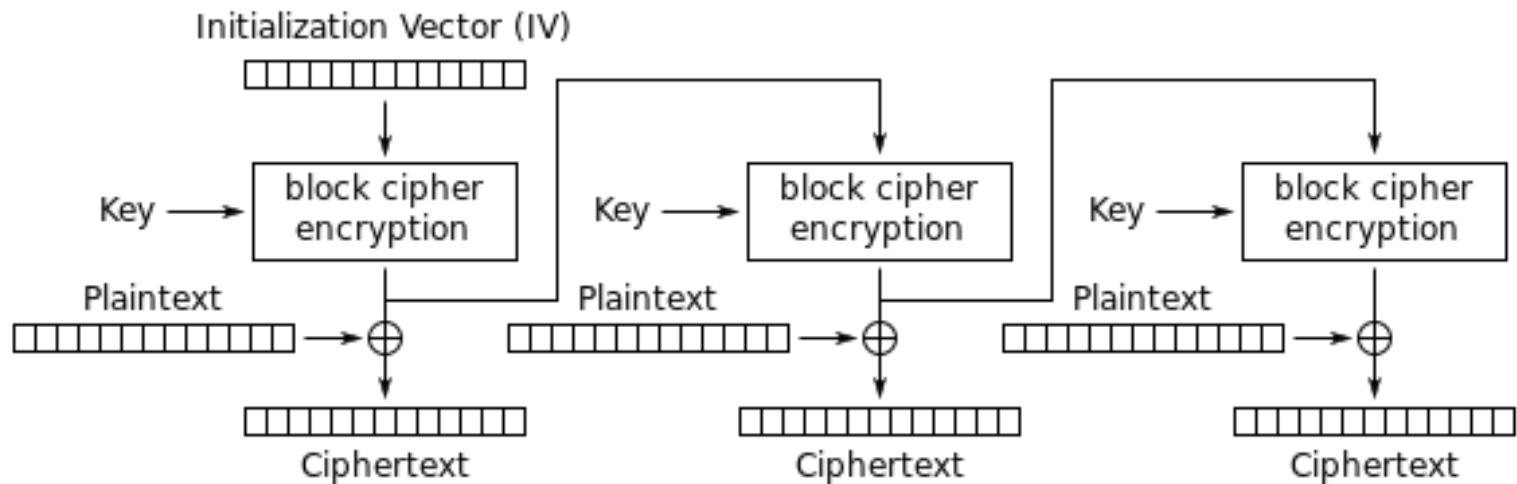
Block Ciphers

- **Cipher text Feedback** – take previous cipher text, encrypt, then XOR with plaintext



Cipher Feedback (CFB) mode encryption

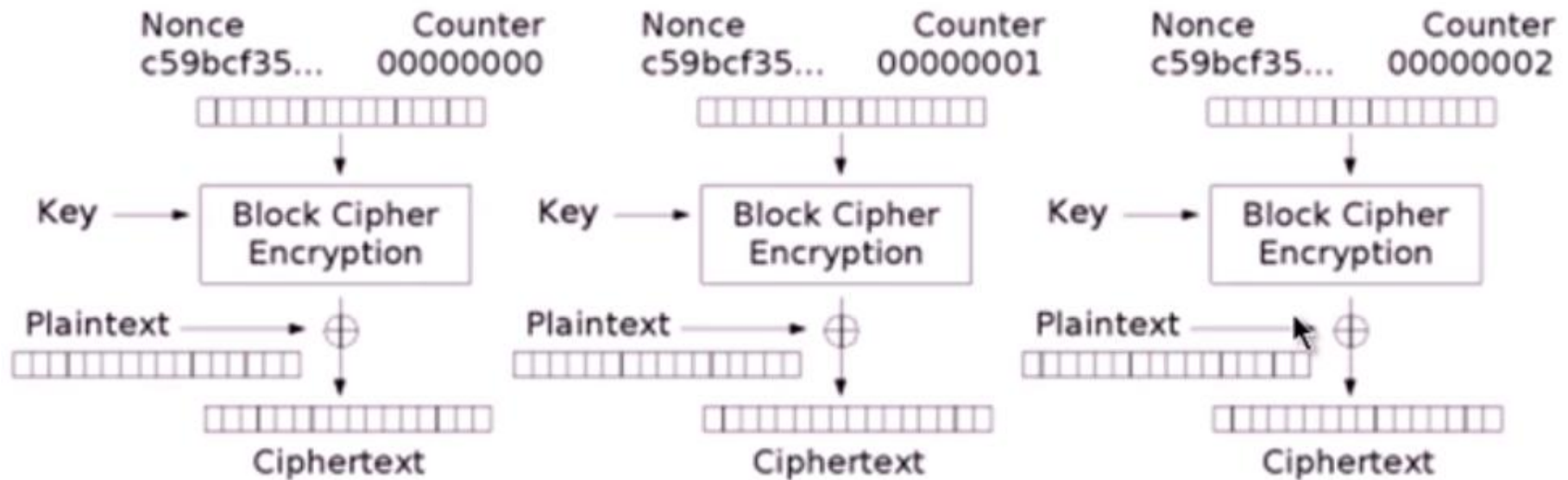
- **Output Feedback** – encrypt previous output, then XOR with plaintext to get cipher text, uses counters to determine where to take from output



Output Feedback (OFB) mode encryption

Counter Mode

- Counter Mode** – A Nonce and an incrementing counter is taken in place of IV and encrypted with key, the cipher text is then XORed with plaintext to get final cipher text



Counter (CTR) mode encryption

Block Ciphers



- Basic Construction:
 - Generate key pair
 - Encrypt plaintext
 - Break message into consecutive blocks of length l (possibly have to augment the last block with some padding)
 - Encrypt each block with encryption key e
 - Decrypt ciphertext
 - Decrypt each block with decryption key d and concatenate blocks less padding to get plaintext

Stream Ciphers (State Cipher)



- A stream cipher is a symmetric key cipher where plaintext bits are combined with a pseudorandom cipher bit key stream, typically by an exclusive-or (xor) operation.
- In a stream cipher the plaintext digits are encrypted one at a time, and the transformation of successive digits varies during the encryption.
- An **alternative name** is a **state cipher**, as the encryption of each digit is dependent on the current state.

Stream Ciphers (State Cipher)



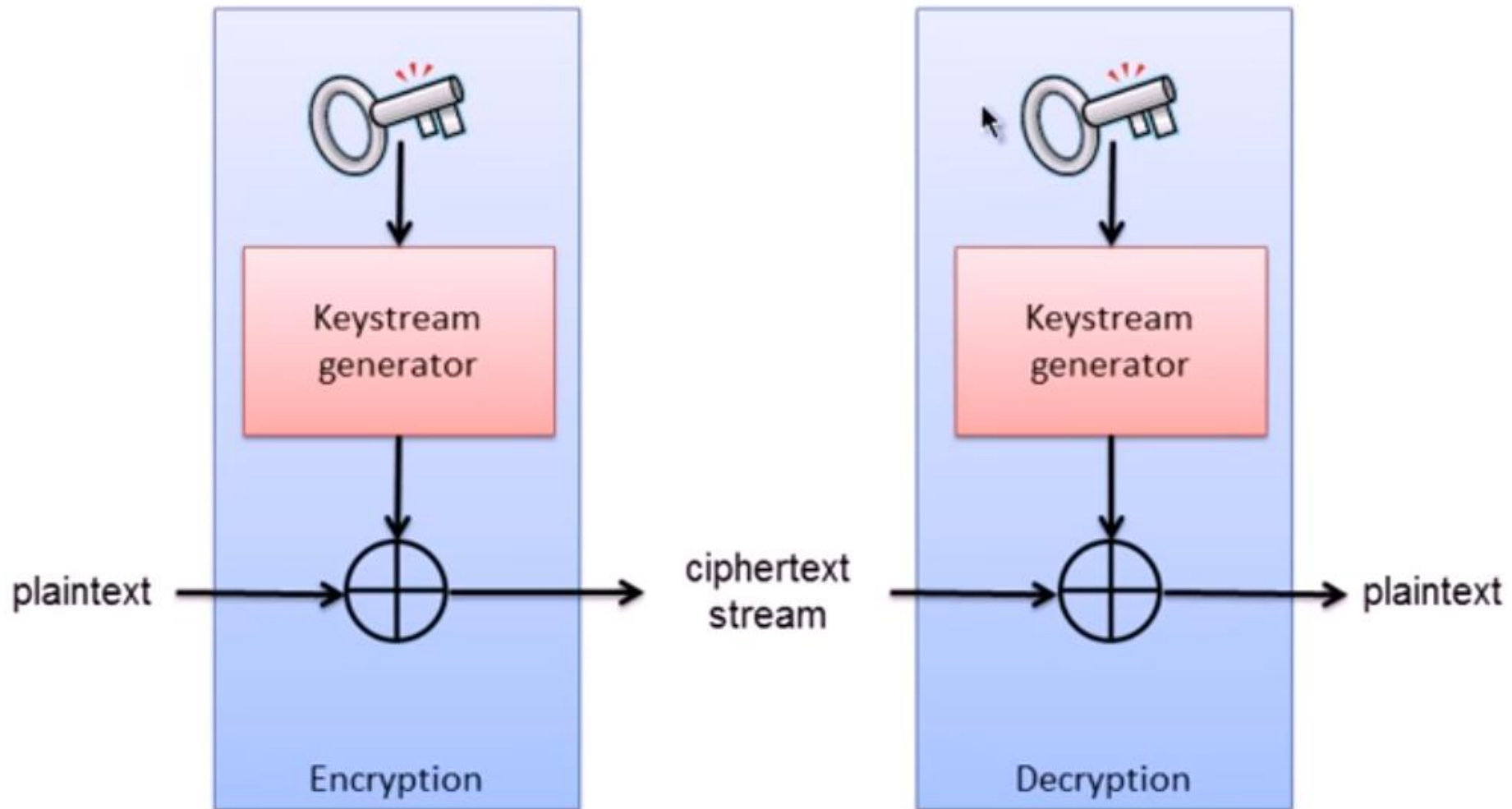
XOR

- Boolean mathematical function which creates an output bit based on two input bits.

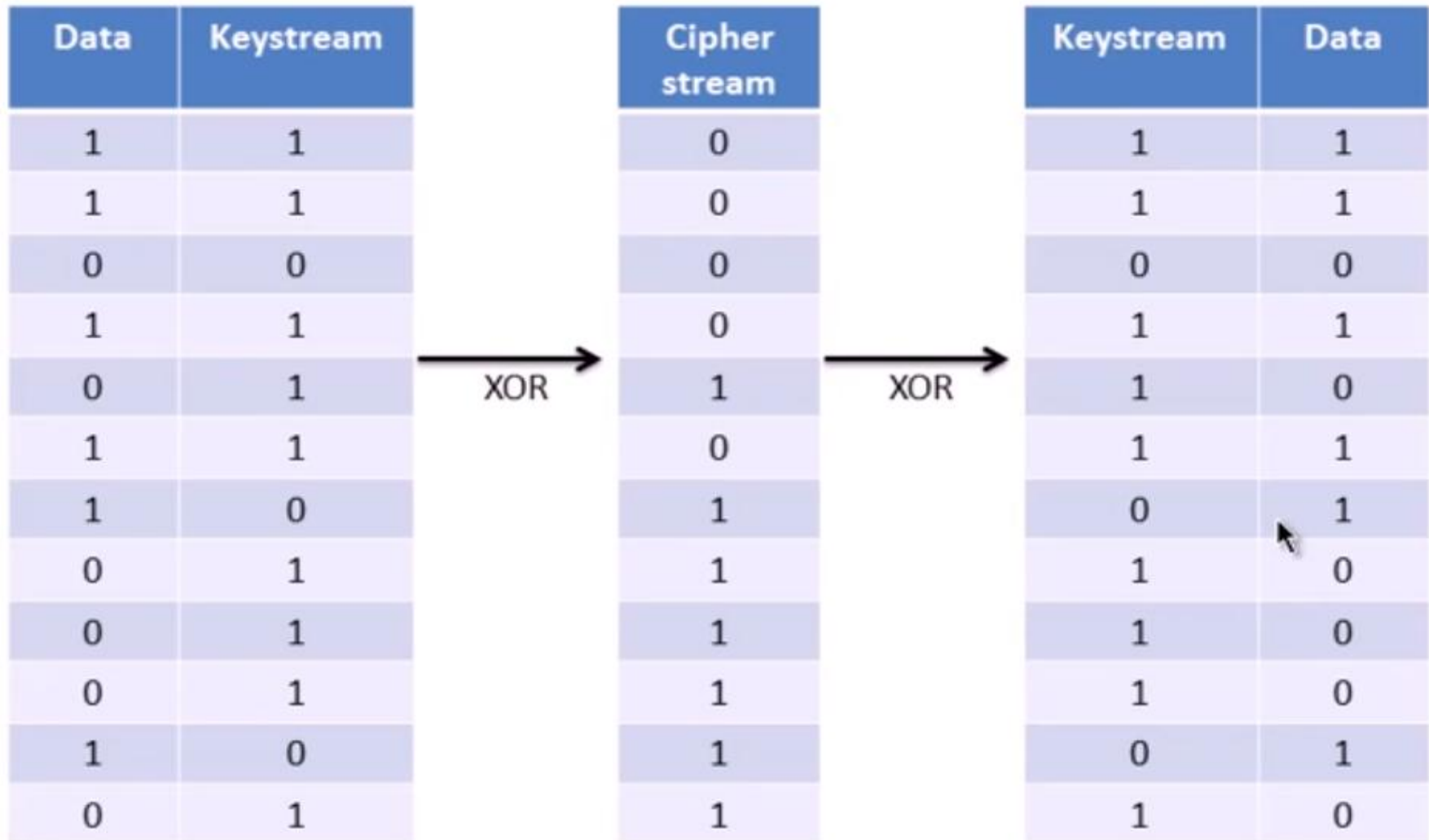
INPUT1	INPUT 2	XOR	OUTPUT
--------	---------	-----	--------

0	0	=	0
0	1	=	1
1	0	=	1
1	1	=	0

Stream Ciphers (State Cipher)



Stream Ciphers (State Cipher)



- **Block Ciphers**

- DES
- AES
- IDEA
- Blowfish
- RC5
- GOST
- CAST
- SAFER etc...

- **Stream Ciphers**

- RC4
- SEAL
- SNOW
- FISH
- Trivium
- VEST etc...

Ex:



- **Block Ciphers :**

- 128-bit block ciphers*

- Twofish
 - Serpent
 - Rijndael

- 64-bit block ciphers*

- IDEA
 - Blowfish
 - DES

- **Stream Ciphers**

- RC4
 - SEAL
 - SNOW

S-Box

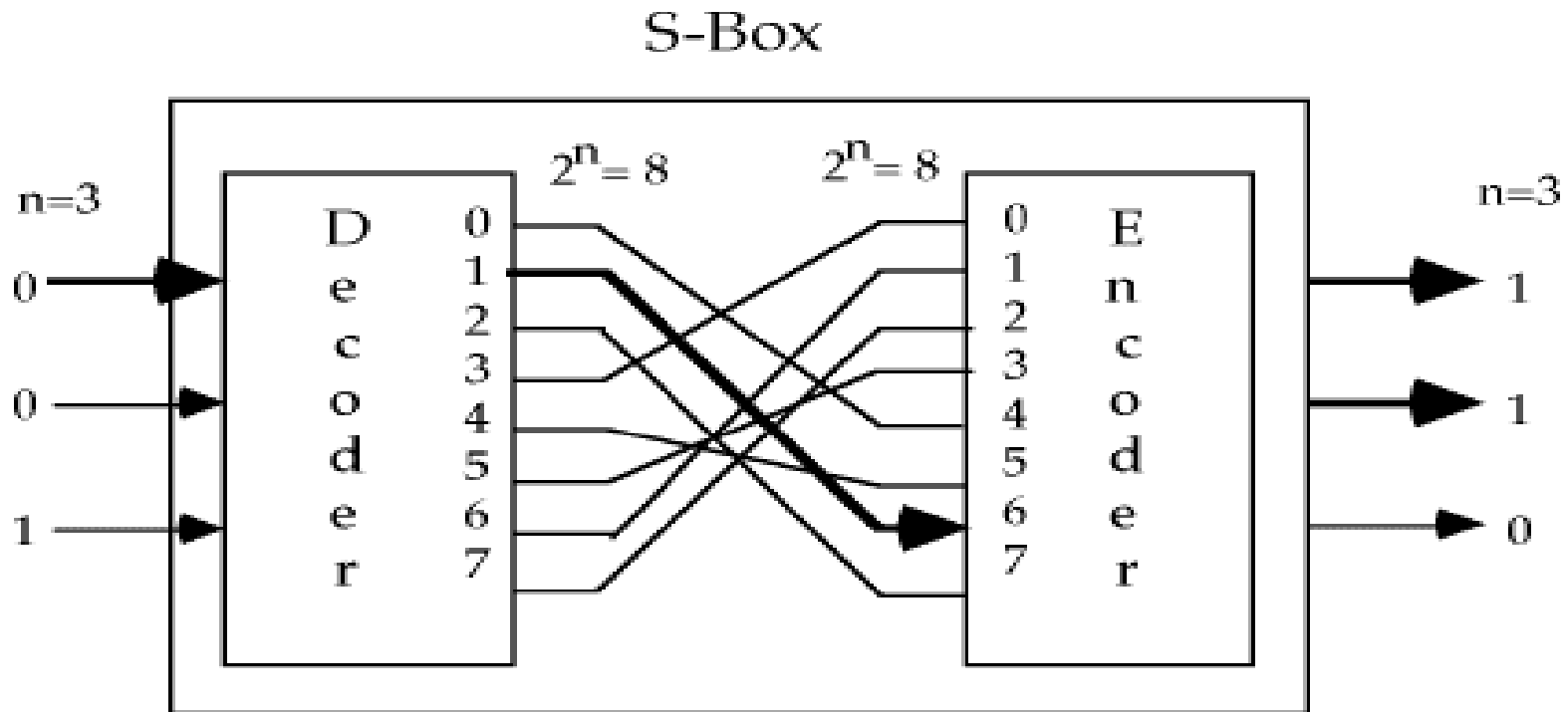


Fig 2.1 Substitution Operation

P-Box

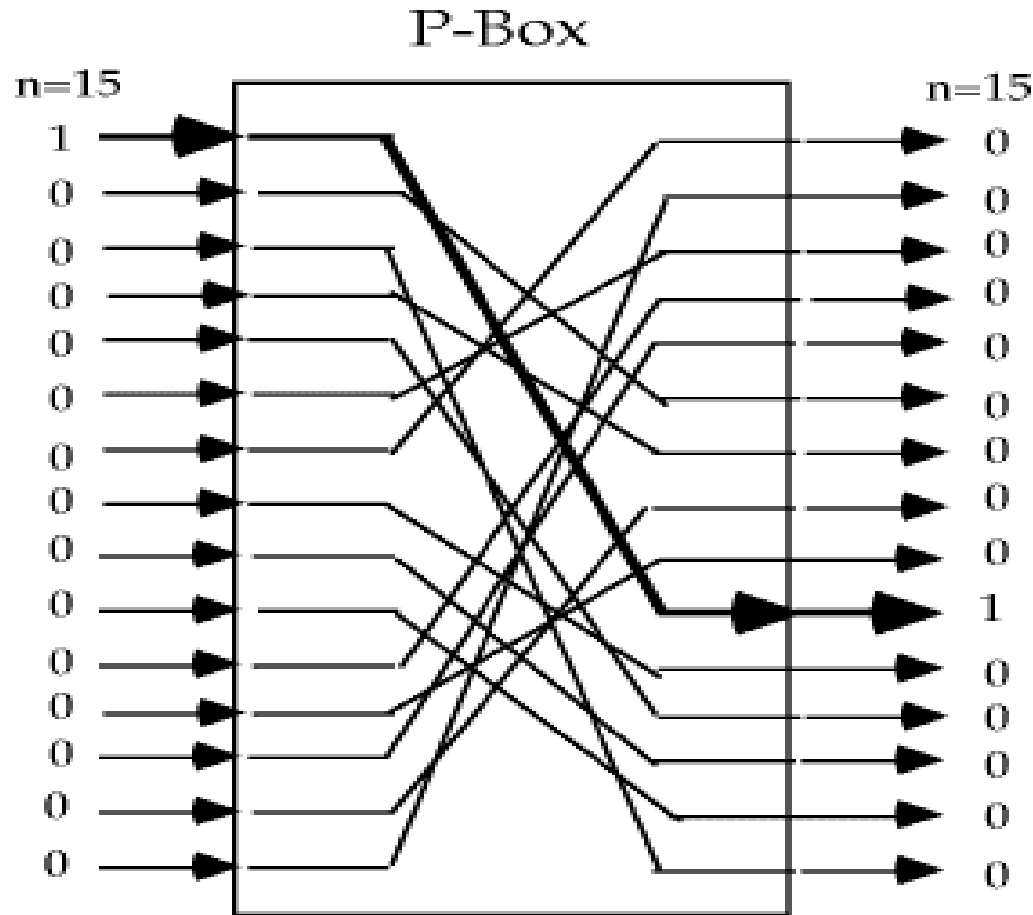


Fig 2.2 - Permutation or Transposition Function

Choice of Crypto Algorithm



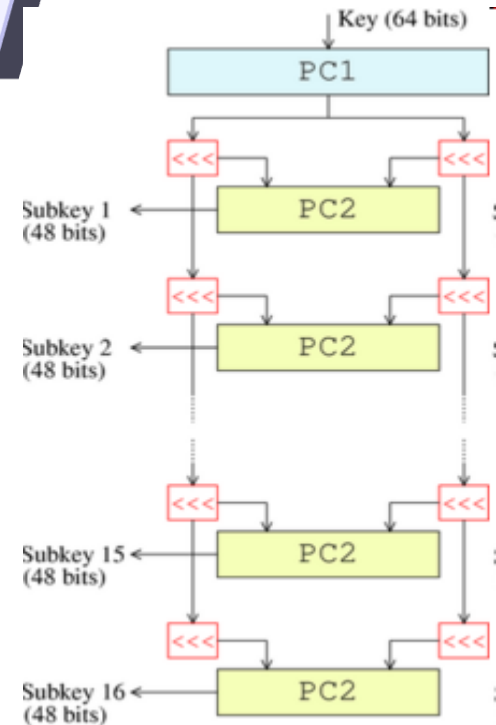
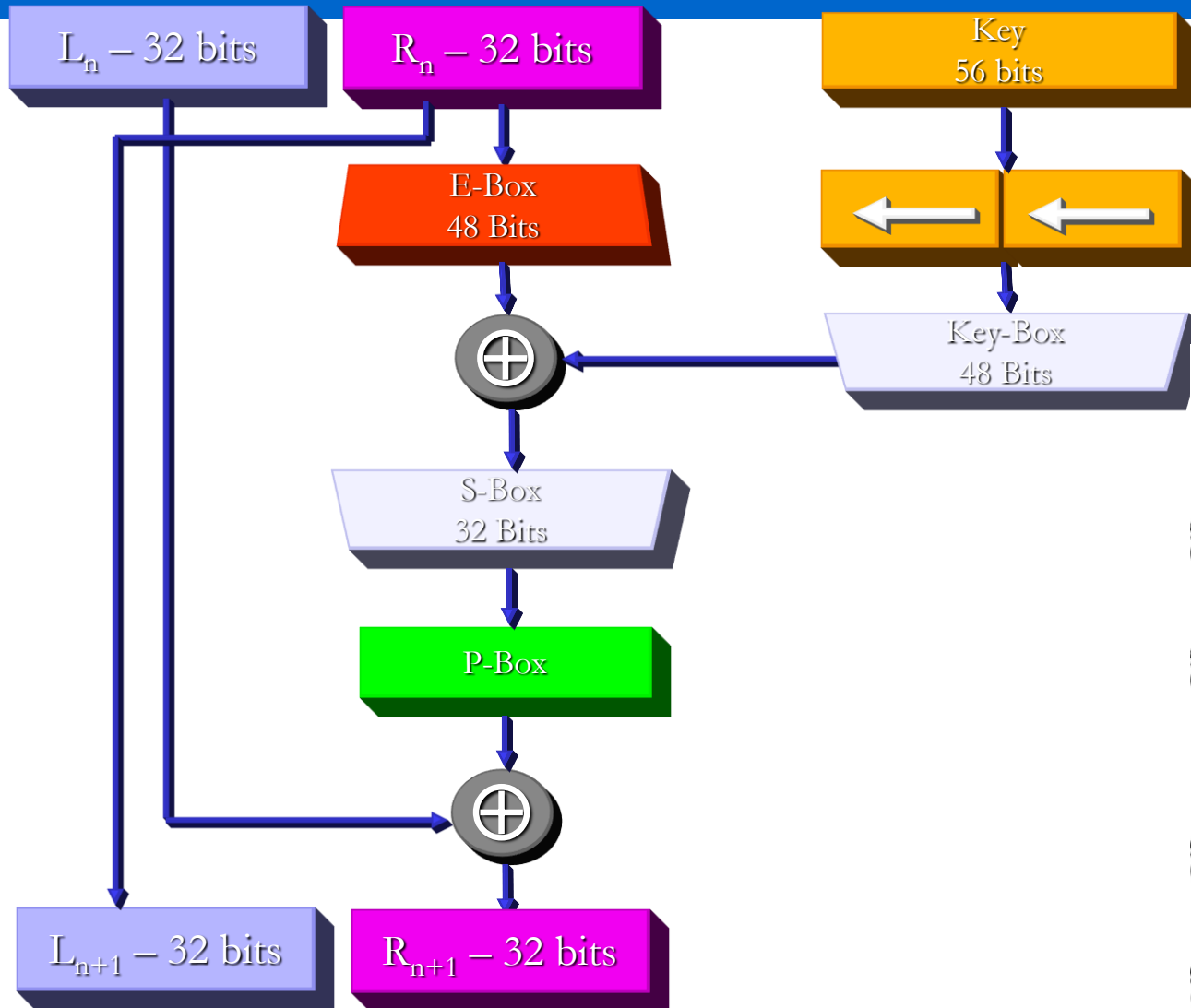
- No. of Rounds
- Combination of S & P Boxes
- Who has implemented the Algorithm
- Whether it is Open Source or Proprietary

Data Encryption Standard



- **DES** (Data Encryption Standard)
 - a very popular encryption scheme
 - Processes blocks of data at a time.
 - A simplified version of DES, i.e S-DES is used for educational purposes.
- Created by IBM called LUCIFER
- Adopted in 1977 by National Bureau of Standards (now NIST)
- 56 bit key to encrypt 64 bit blocks
- Consists of 16 stages plus initial/final permutations

DES – One Round



- Weak key size
 - Originally used a 128 bit key
 - Shortened to 56 bits to fit on 1 chip
- Brute force attacks
 - Deep Crack– EFF built \$210K system
 - Distributed.Net– 1000s of Internet connected systems working together

Hash Function



- A hash function is a cryptographic mechanism that operates as one-way function
 - Creates a digital representation or "fingerprint" (Message Digest)
 - Fixed size output
 - Change to a message produces different digest

Examples : MD5 , Secure Hashing Algorithm (SHA)

Hash function - Properties



Consistency

-  Same input must produce the same message digest. No randomness

Uniqueness

-  Computationally infeasible to identify two messages that will generate the same message digest

One way

-  Computationally infeasible to identify the input given the message digest

Hash - Example

Message

Hi Jai,
I will be in the park at
3 pm
Veeru

Hi Jai,
I will be in the park at
8 pm
Veeru

Hash Algorithm

Message Digest

cfa2ce53017030315fde705b9382d9f4

d4216ytf6b9385fe502b165dfe8cec17

Digests are Different

MD5 and SHA

Message

Hi Jai,
I will be in the
park at 3 pm
Veeru

MD5

Message Digest

cfa2ce53017030315f
de705b9382d9f4

128 Bits

Hi Jai,
I will be in the
park at 3 pm
Veeru

SHA-1

1f695127f210144329ef
98e6da4f4adb92c5f18
2

160 Bits

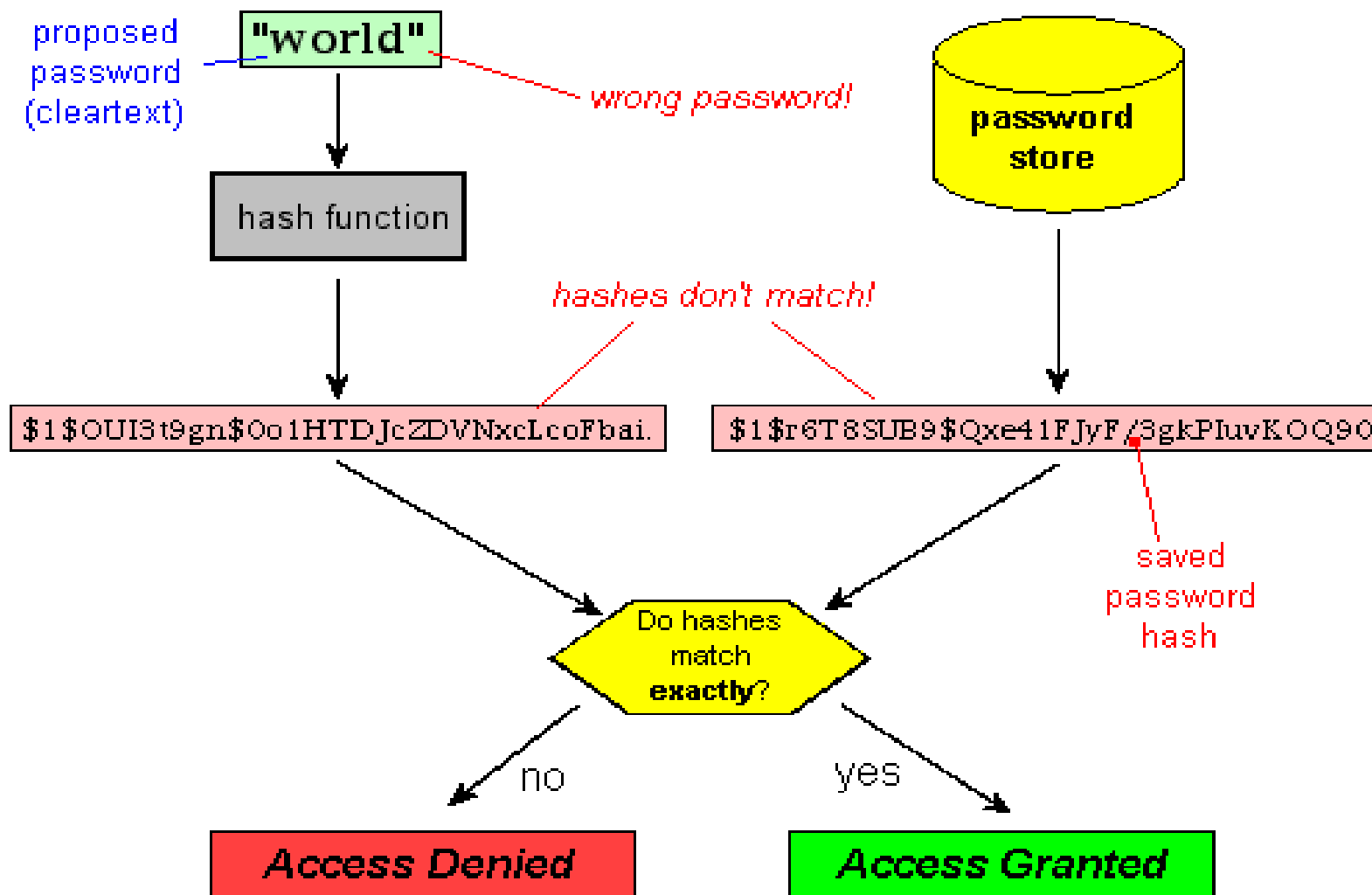
Hi Jai,
I will be in the
park at 3 pm
Veeru

SHA-2

2g5487f56r4etert654tr
c5d5e8d5ex5gttahy55e

224/256/384/512

Example of Hash functions



Symmetric Key Cryptography



- Also called as Secret Key Cryptography or Single Key Cryptography.
- Uses one key shared by both sender and receiver.
- This key is used for both encryption and decryption.
- Both parties have to agree on the key before start of the communication
- Encryption and Decryption is extremely fast comparing to asymmetric cryptography

Number of keys required

For 'n' number of communicating parties the no. of keys required will be

$$\# \text{ of Keys} = (n(n-1))/2$$

Eg: $n = 5$

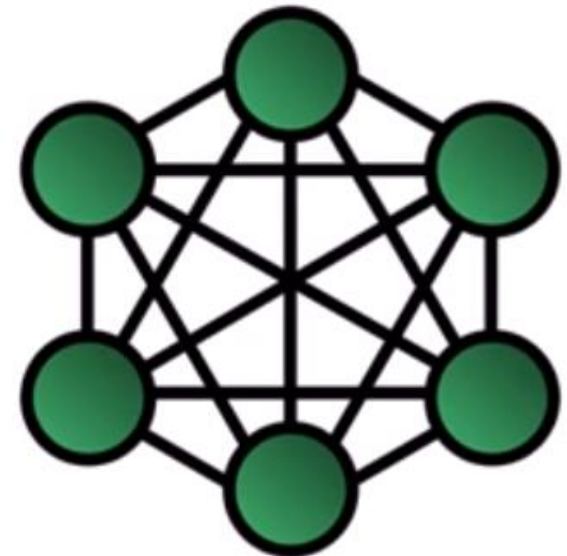
$$\text{keys} = (5(4))/2 = 10$$

$n=10$

$$\text{keys} = (10(9))/2 = 45$$

$n=100$

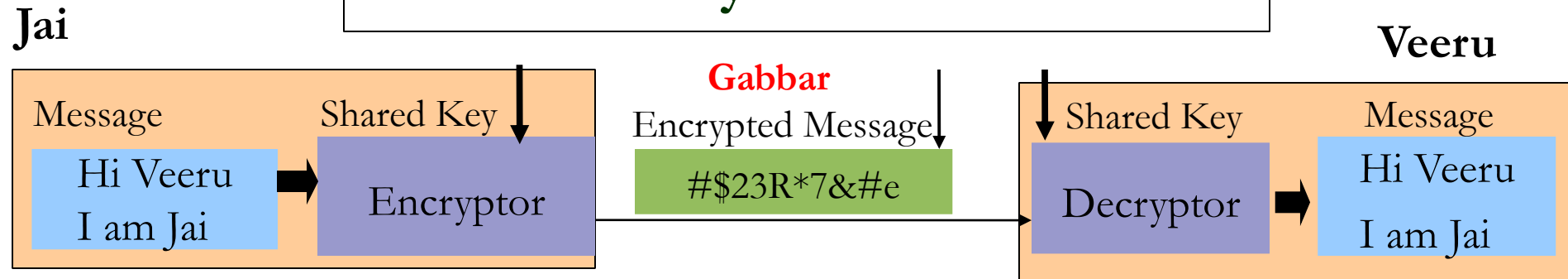
$$\text{keys} = (100(99))/2 = 4950$$



Symmetric Key Cryptography



Confidentiality & Authentication



Unauthorized Login Attempt



Symmetric Key Cryptography

Integrity

Jai

Veeru

Gabbar

Enc. Digest

Message

Shared Key

Hi Veeru

I am Jai

Digest

Message

Hi Veeru I am Jai

%*t%

Shared Key

Enc. Digest

Dec. Digest

Message

Comp. Digest

Confidentiality & Integrity

Shared Key - 2

Shared Key - 2

Veeru

Message

Enc. Digest

Hi Veeru

I am Jai

%*t%

Gabbar

Encrypted Message

#\$23R*7&#e

#\$23R*7&#e

Enc. Digest

Message

Issues:

- Jai and Veeru must agree on the secret key without anyone else finding out
- Compromise of shared key leads to compromise of communication
- Secure Key Distribution and Scaling

What can be achieved using Symmetric Key ?

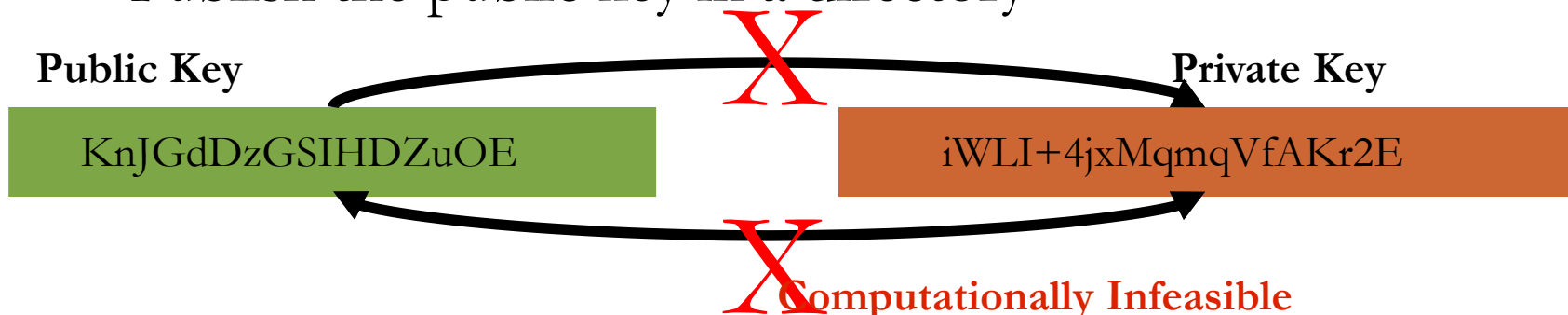
- Confidentiality
- Integrity
- Authentication

What about Non-repudiation ?

Asymmetric Key Cryptography



- Also called as Public Key Cryptography
- Uses a related key pair wherein one is Private key and another is Public key
 - One for encryption, another for decryption
- Knowledge of the *encryption* key doesn't give you knowledge of the *decryption* key
- A tool generates a related key pair (public & private key)
 - Publish the public key in a directory



Asymmetric Key Encryption



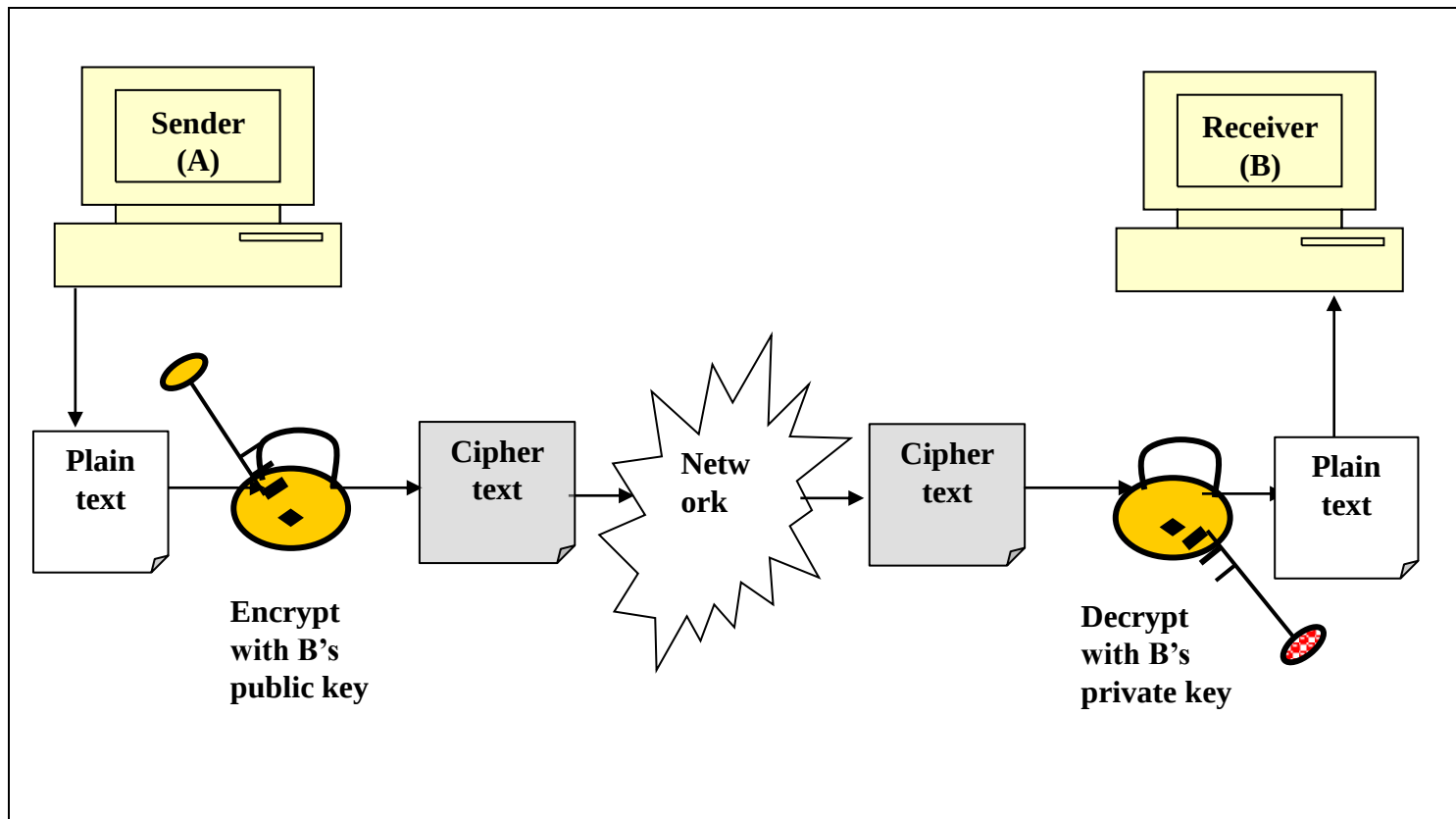
- Important to know who should know which key(s)
- In general:
 - Sender encrypts with recipient's public key
 - Recipient decrypts with its private key

Matrix of Keys



Key details	A should know	B should know
A's private key	Yes	No
A's public key	Yes	Yes
B's private key	No	Yes
B's public key	Yes	Yes

Asymmetric Key Cryptography

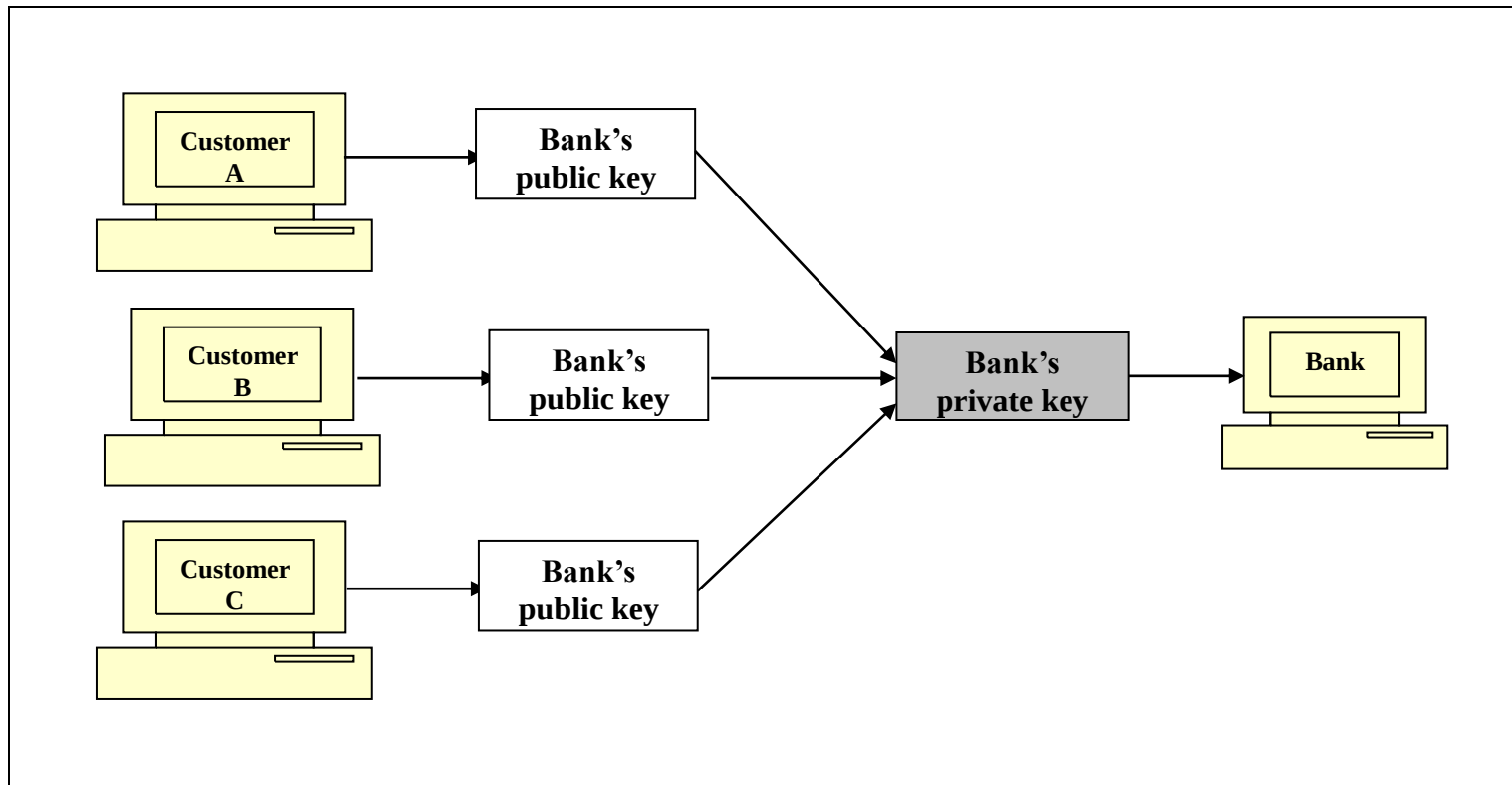


Asymmetric Key Example



- Consider a bank and its customers
- Customers encrypt their messages with bank's public key
- Bank decrypts messages with its private key

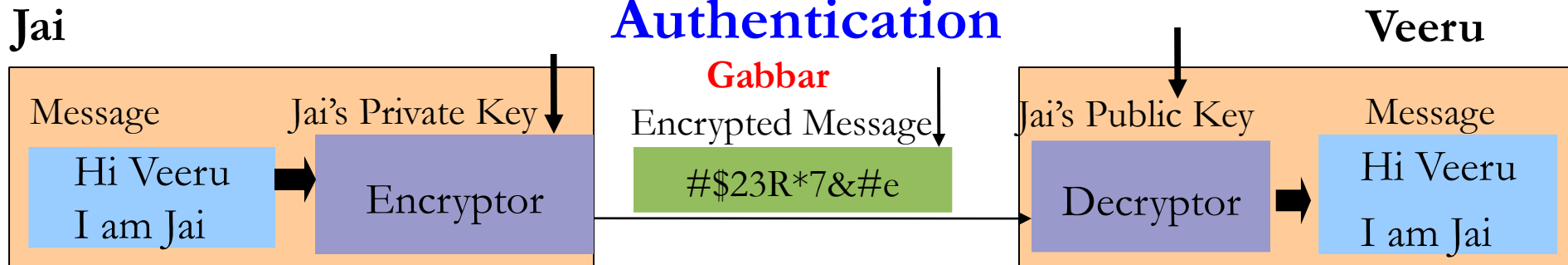
Asymmetric Key Cryptography -- Example



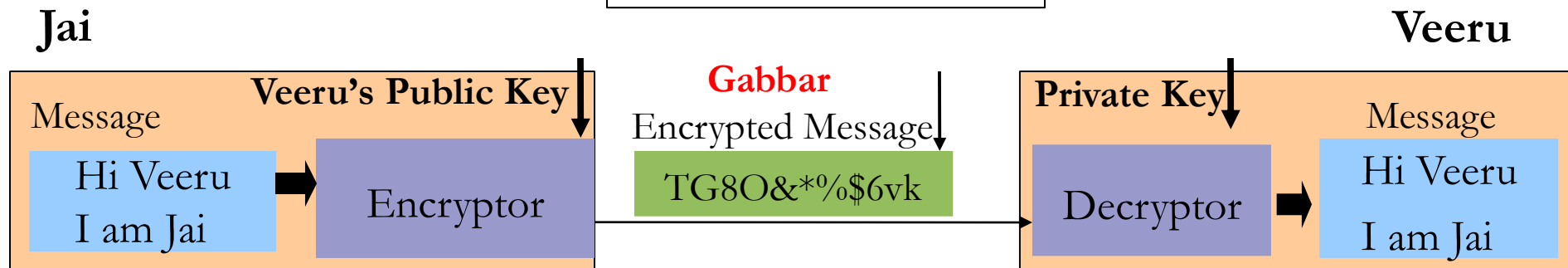
Asymmetric Key Cryptography



Authentication



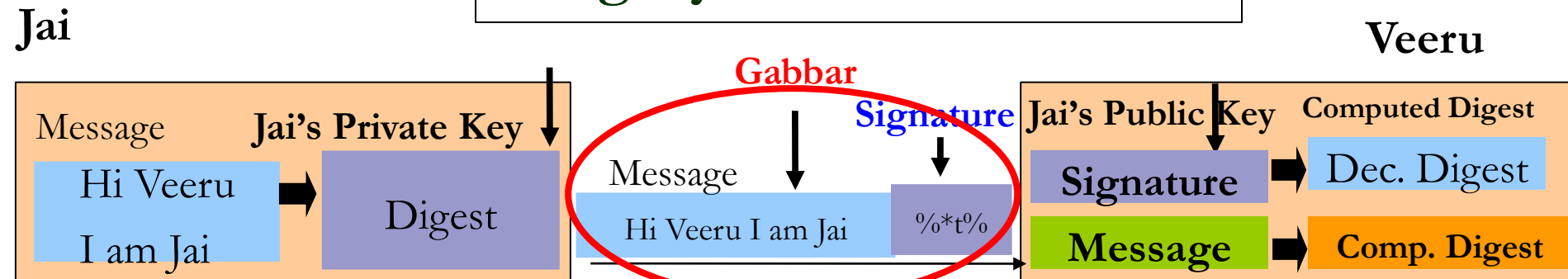
Confidentiality



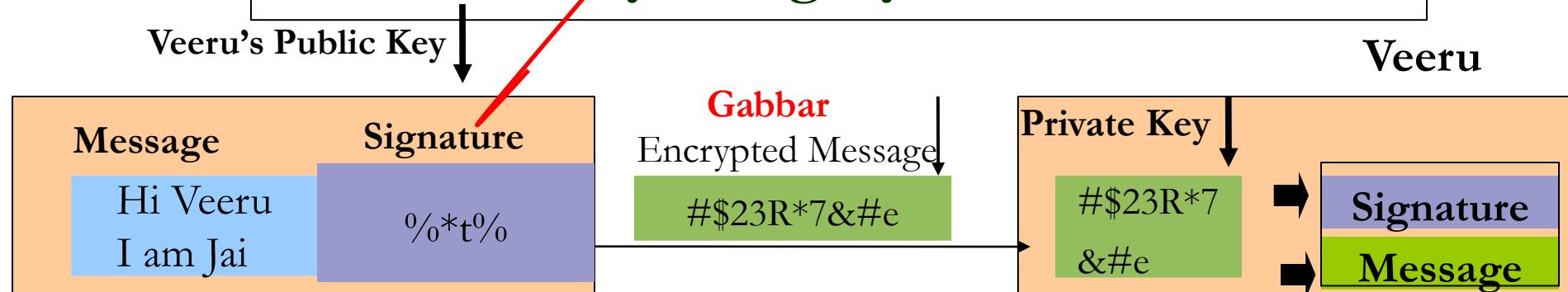
Asymmetric Key



Integrity & Authentication



Confidentiality, Integrity & Authentication



Weakness

- Extremely slow

Strength

- Solves problem of passing the key

Key Aspects

- Public key encryption; RSA

Misconceptions

- More secure
- Has made Symmetric encryption obsolete

Example Public Key



mein-key - WordPad

Datei Bearbeiten Ansicht Einfügen Format ?

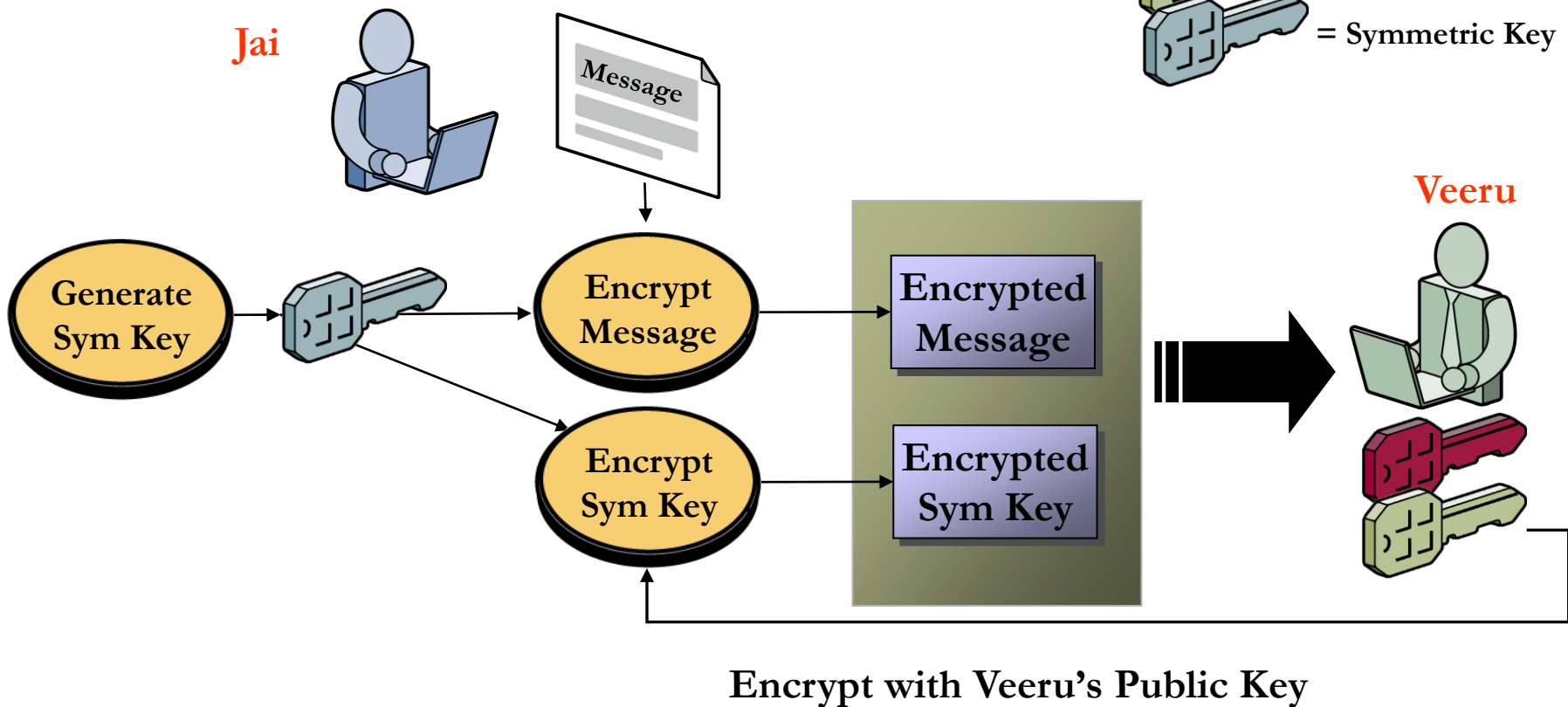
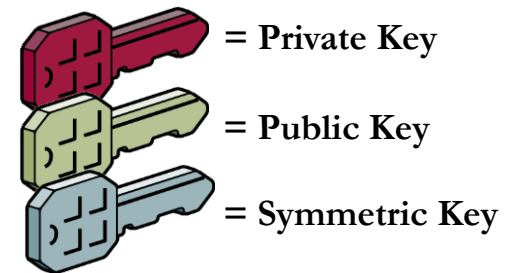
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-----END PGP PUBLIC KEY BLOCK-----

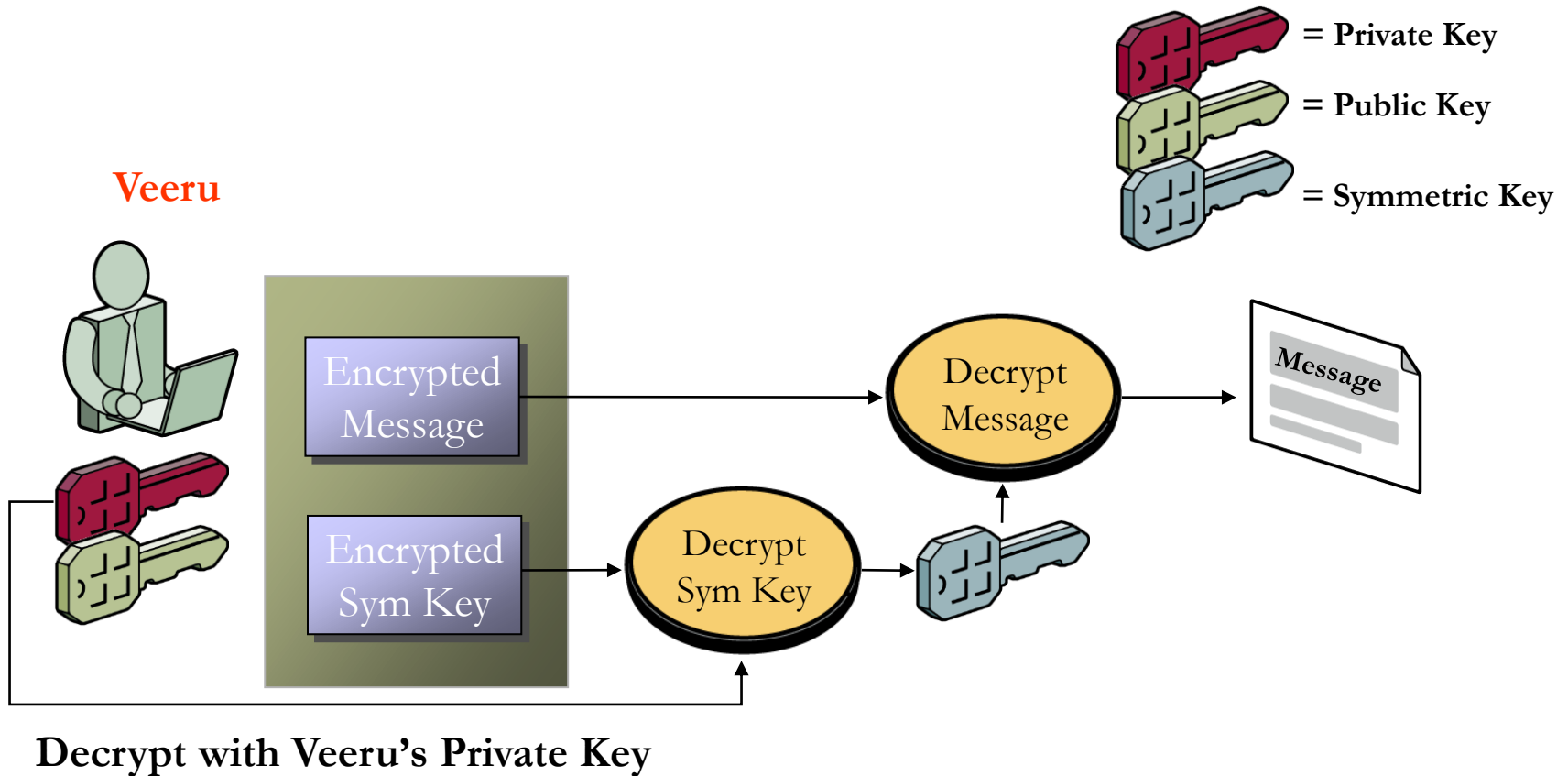
Drücken Sie F1, um die Hilfe aufzurufen.

Public Key Encryption

Symmetric keys encrypt data;
Public keys encrypt symmetric keys



Public-Key – Decryption



**Public key and symmetric key cryptography
are complementary technologies**

References



- Cryptography and Network security – principles and practice :
William Stallings
- Applied Cryptography, Second Edition: Bruce Schneier
- www.certicom.com/index.php/ecc-tutorial
- http://campustechnology.com/articles/39190_2
- <http://csrc.nist.gov/>
- Handbook of Applied Cryptography, by Menezes
- <http://en.wikipedia.org>
- Cryptographic Techniques for N/w Security

Thank You